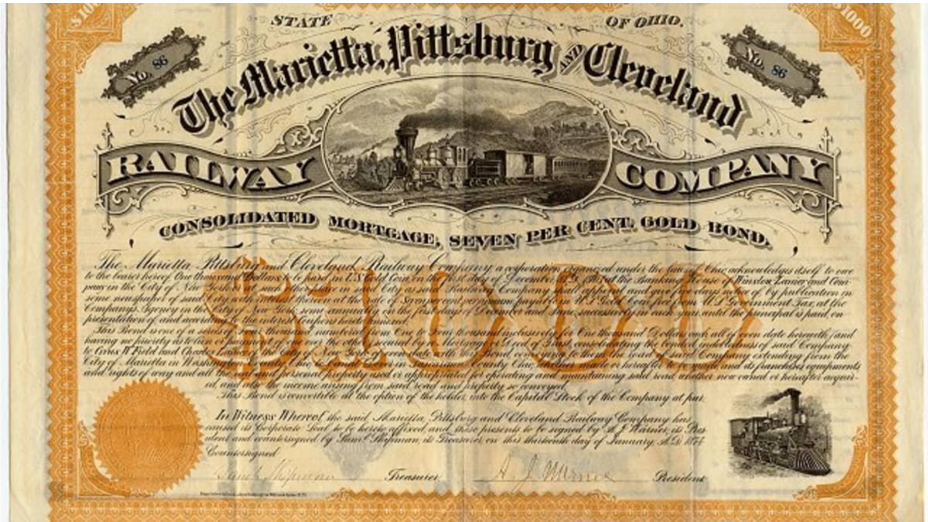
Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
Fall 2025

Prof. Francesco Bripi

The Bond Market

(Chapter 12)



Chapter Preview

- ✓ Last time we studied Money markets
- ✓ We now focus on long-term bonds with >1Y maturity
- ✓ **Capital market** = Bond market + Stock market
- ✓ Firms raise **capital** directly in these markets for *long-term* financing and capital projects (*investment*)

Capital Market Participants

- ✓ Primary **issuers** of securities in capital markets
  - *governments* (Treasury, Agencies, Municipalities): debt only
  - *corporations*: equity and debt
- ✓ Largest **buyers** of securities
  - *households* often through financial intermediaries (banks, mutual funds, pension funds, etc …)
- ✓ The choice between debt and equity in corporations is called <u>capital structure</u>
  - many strategic reasons other than raising capital to get listed on stock market
  - but issuing bonds certainly to satisfy financing needs

Bond market

In comparison with **Money Market:**

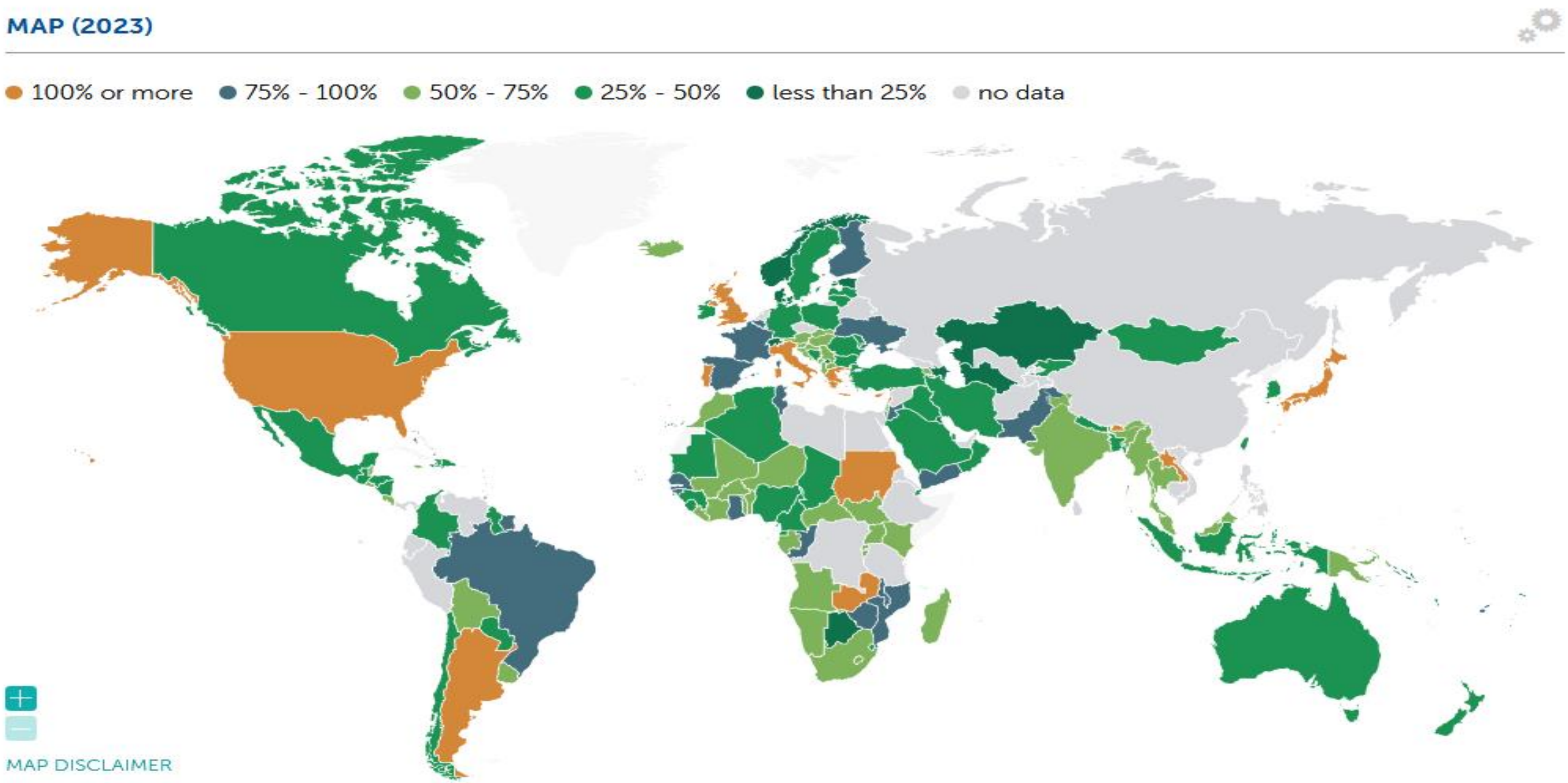
- ✓ bond market instruments *maturity*:
  - ○ ≥ 1 year from original issue date
    - ▪ medium-term (≥1 year and <10 years)
    - ▪ long-term (≥10 years) debt instruments
  - ○ bond market is then a capital market
- ✓ but trade of longer term debt instruments …
  - ○ … implies higher default risk (and interest rate risk)
- ✓ bonds play a more prominent role in investment decisions
  - ○ trade mainly in OE's (Organized Exchanges)
  - ○ differently money mkts instruments mainly traded in OTCs

Bonds

Today we will be looking at:

1. **Government bonds**, especially US Treasuries
2. **Municipal bonds**, issued by US states/cities
3. **Agency bonds**, issued by quasi-public entities
4. **Corporate bonds**, issued by corporates
5. **Bond Pricing**: semi-annual Coupon

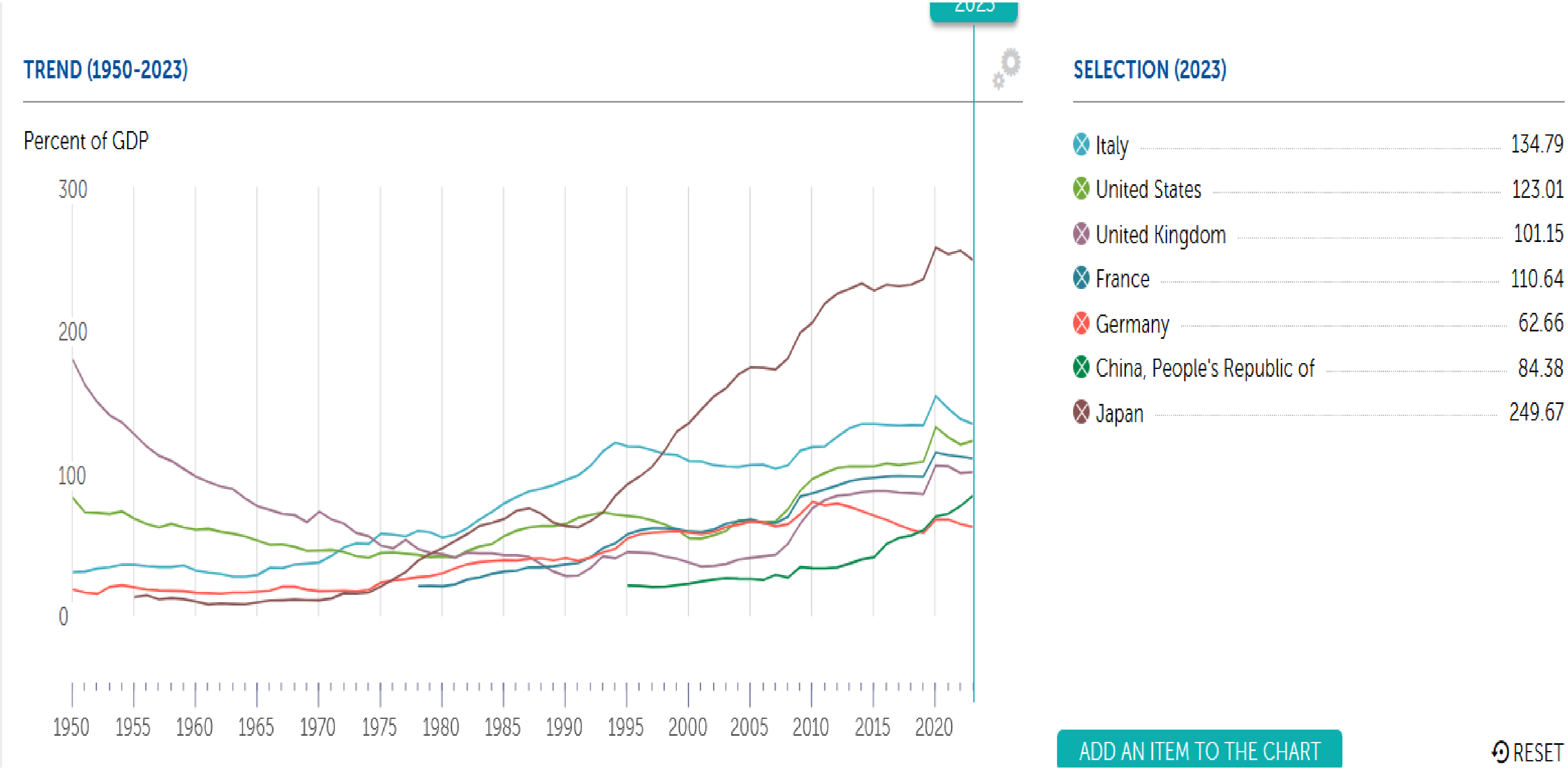
Government debt (as % of GDP): in the world (but no Russia, nor China)



Source: IMF, Global Debt database

Government bonds are important but their **risk** and **price** may change across countries

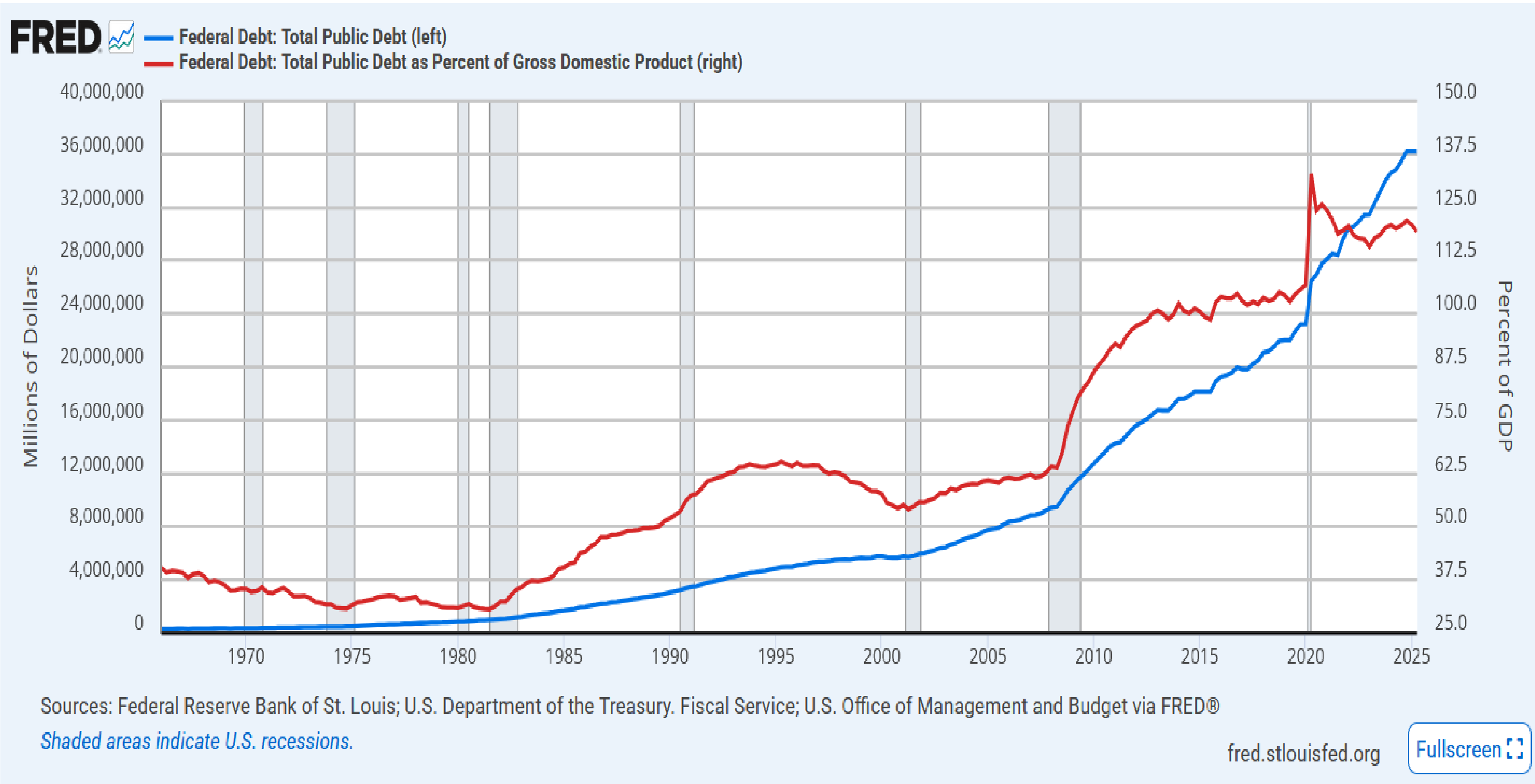
Government debt (as % of GDP): main economies



Source: IMF, [Global Debt database]

Government bonds matter because *all governments need external financing*!

US Federal Debt



Growth of US pulic debt, both in value and as a % of GDP

Treasury Bills, Notes and Bonds: maturity

US Treasury debt

Money market

Treasury bill (≤1 year)

Bonds market

Treasury note (1 to 10 years)

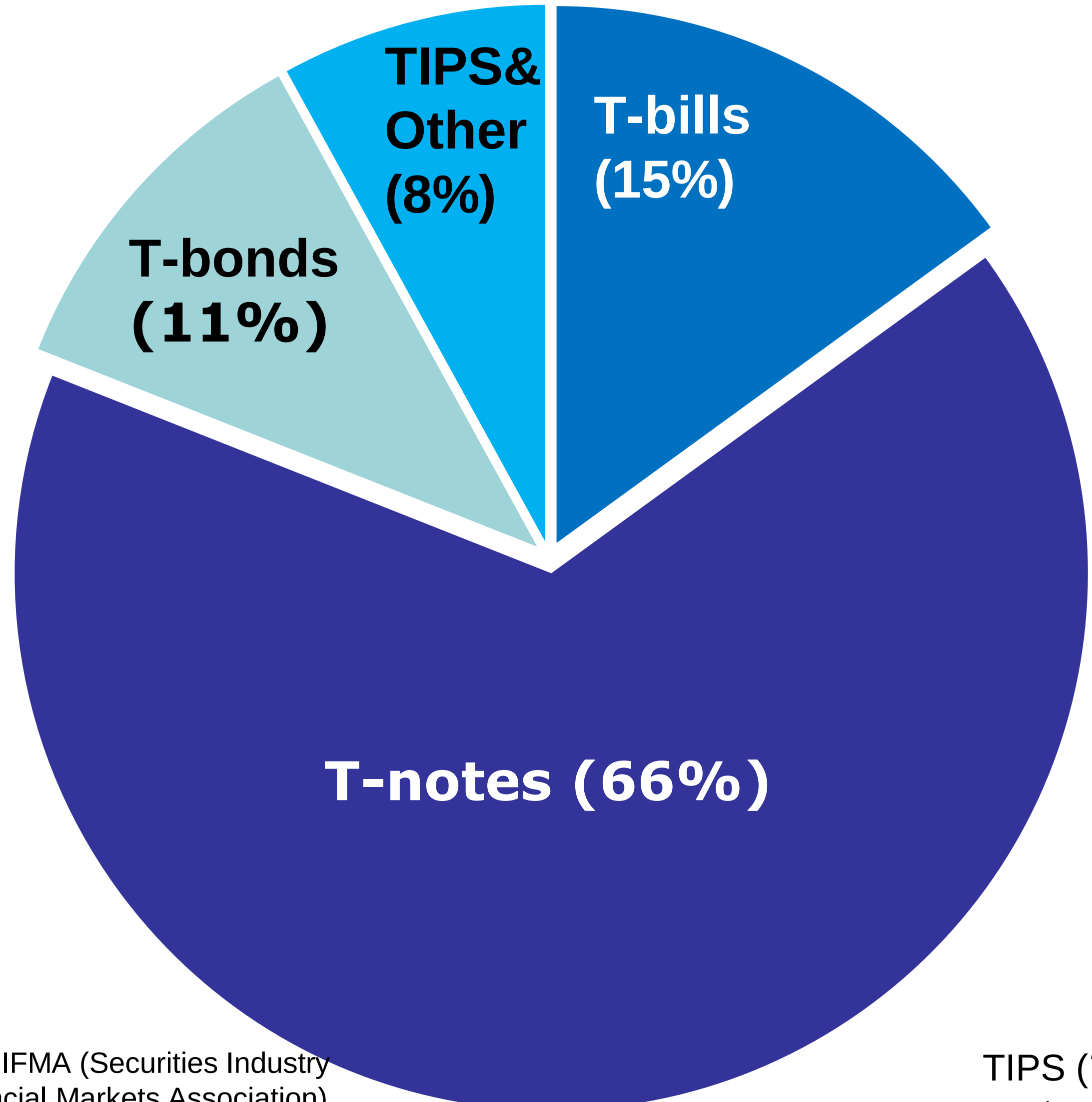
Treasury bonds (>10 years)

Type	Maturity
Treasury bill	Less than 1 year
Treasury note	1 to 10 years
Treasury bond	10 to 30 years

✓ Typical maturities:

- Treasury notes: 2, 3, 5, 7, 10 years
- Treasury bonds: 20, 30 years

Treasury Bond Market by Type of Bonds



Source: SIFMA (Securities Industry and Financial Markets Association)

TIPS (***T****reasury* ***I****nflation* ***P****rotected* ***S****ecurities*)

- type of Treasury bond where principal and interest are indexed to inflationary measure

Government debt, bond risk and interest rates

- ✓ Risk on gov bonds is assessed by <u>credit-rating agencies</u> (CRAs). Largest are *<u>Moody's, S&P, Fitch, DBRS</u>*
  - US government bonds are considered a *risk-free* asset ⟹ practically <u>no default risk</u>
  - but recently downgraded
  - the T-bills are even "more" risk-free (wrt and T-notes and T-bonds) because of shorter maturity
- ✓ Interest rate correlated with rating
  - interest rates on T-notes and T-bonds are generally low
  - but for some countries yield may be high (Italy and Greece!)
- ✓ Not only default risk… of course there is still inflation risk (and do not forget IRR!)
- ✓ Optional: who owns US public debt?

US Municipal Bonds

- ✓ **Municipal bonds**
  - o issued by local authority: county or state governments

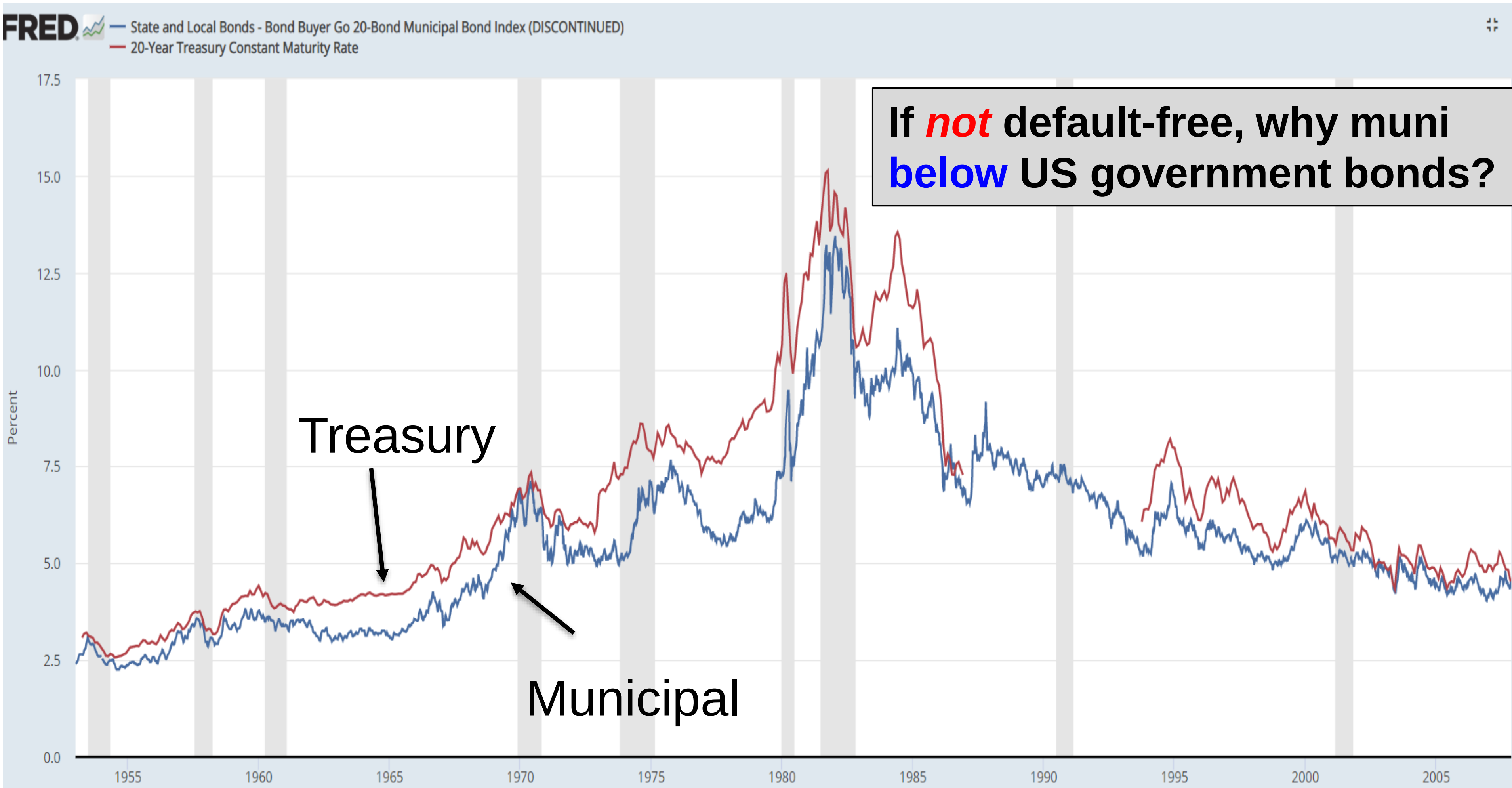
- ✓ Two types:

1. **Revenue Bonds**
   - backed by the cash flow of a <u>particular</u> <u>project</u> (bridge, toll road)

2. **General Obligation Bonds**
   - do <u>not</u> have a <u>specific</u> <u>project</u>
   - backed by "full faith and credit" of the local government

- ✓ **Muni bonds are *NOT*** *default free*
  - o some examples:
    - Orange County, Cal: 1994 losses $1.4bn
    - WHOOPS, DC: 1983, $2.25bn
    - Jefferson Cnty, AL: 2011 sewer revenue bonds 3.7bn
    - Detroit, MI: 2013 $18.5bn, biggest ever

Municipal vs Treasury comparison, 1955-2008



Municipal Bonds: Tax exemption

Because they are **tax-free**!

Q: Suppose the rate on a *corporate* bond is 9% and that on *municipal* bond is 6.75%. Which one should you choose?

Municipal Bonds: Tax exemption

Because they are **tax-free**!

Q: Suppose the rate on a *corporate* bond is 9% and that on *municipal* bond is 6.75%. Which one should you choose?

A: It depends on the Marginal Tax Rate (MTR)!

Municipal Bonds: Tax exemption

Because they are **tax-free**!

Q: Suppose the rate on a *corporate* bond is 9% and that on *municipal* bond is 6.75%. Which one should you choose?

A: It depends on the Marginal Tax Rate (MTR)!

To correctly compare another bond with a muni bond interest rate, must find the Marginal Tax Rate that makes you indifferent between the two:

$$\textit{Muni rate} = \textit{Other\_bond rate} \times (1 - \textit{Marginal Tax Rate})$$

$$i_{Mun} = i_{Oth}(1 - MTR)$$

$$\frac{i_{Mun}}{i_{Oth}} = 1 - MTR \Rightarrow \boldsymbol{MTR = 1 - \frac{i_{Mun}}{i_{Oth}}}$$

Municipal Bonds: Tax exemption

Plug in data from previous example to find the MTR:

$$i_{Mun} = i_{Oth}(1 - MTR) \Rightarrow 0.0675 = 0.09 \cdot (1 - MTR)$$

Solution: **MTR* = 25%**

- If Marginal Tax Rate above/below 25%, the municipal bond offers a higher/lower after-tax cash flow …

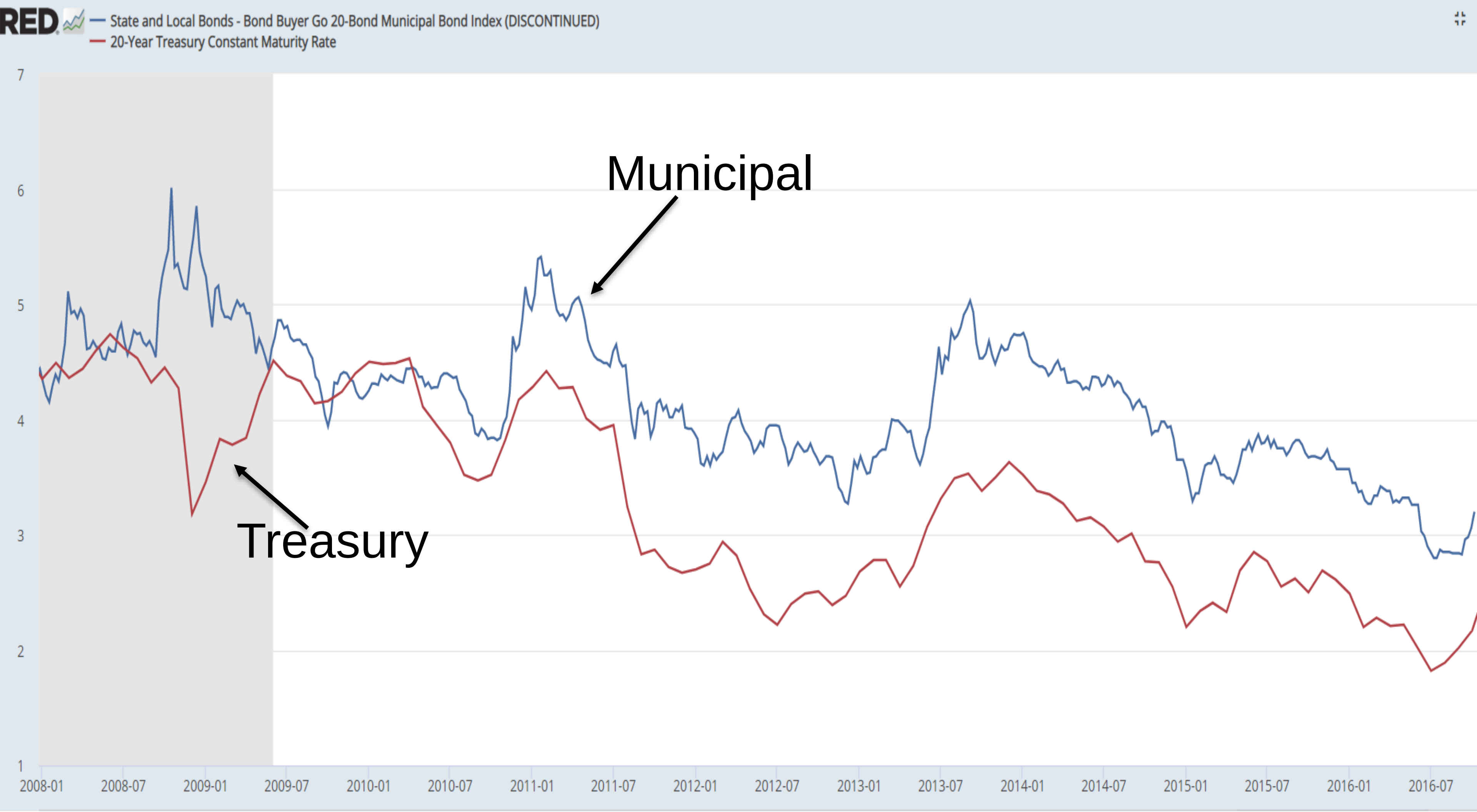
✓ Case 1. Tax rate at **20%** $(MTR < MTR^*)$:

- Corporate bond: $i_{CB} = (1 - 0.2) \times 0.09 = 0.0720 \ (7.2\%)$
- Municipal bond: $i_{Mun} = (1 - 0.0) \times 0.0675 = 0.0675 \ (6.75\%)$

✓ Case 2. Tax rate at **30%** $(MTR > MTR^*)$:

- Corporate bond: $i_{CB} = (1 - 0.3) \times 0.09 = 0.0630 \ (6.3\%)$
- Municipal bond: $i_{Mun} = (1 - 0.0) \times 0.0675 = 0.0675 \ (6.75\%)$

Not so much anymore after 2008: 2008-2016 … why?



Agency Debt

Agency bonds

- ✓ they are issued by US Government-Sponsored Enterprises (GSE)

- ✓ GSEs are private companies (listed on NYSE) that can issue "*government securities*" – *agency bonds*

- ✓ with the proceeds of bonds, GSEs
  - buy *mortgages* from banks and sell them as mortgage-backed securities
  - extend specific loans (e.g.: *student* loans)
  - sell *insurance*

- ✓ The GSE acted with an **implicit guarantee** that the U.S. government would not let them default …
  - …. but also increase ***moral hazard***!
  - see the case of Freddy Mac and Fannie Mae in 2008, world's biggest financial bail-out! Since 2008 under government conservatorship
  - optional: read here … it dates back to 2005!

Corporate Bonds

- ✓ Bonds issued by (largest) corporations
  - remember: only a handful (about 1,000) of companies have access to bond market in the US!
  - some countries adopt easier access criteria for SME's (optional: Mini bonds in Italy, but also Japan, etc.…)

- ✓ *Default risk* is significantly higher than for Treasury bonds, but a lot of heterogeneity between firms

- ✓ Risk is assessed by credit-rating agencies (CRAs)
  - the three largest are: *Moody's, S&P, Fitch*
  - adopt somewhat different scales, but in general:
    - from AAA (low risk) to BBB (high risk): "*investment grade*"
    - from BB to C,D: "*speculative/junk/high yield*"

Who Wants to Buy a Tesla Bond?

Deal's proposed yield of 5.25% is low considering company's junk rating

WSJ, Aug 9 2017

Characteristics of Corporate Bonds

✓ Bonds have a **bond indenture**:

- a contract stating lender's rights and borrower's obligations

For example:

1. **Restrictive covenants**
2. **Call provisions**
3. **Conversion option**
4. **Seniority** and **Collateral** (if any)

Not all corporate bonds have these characteristics.

May be none, may be all.

Characteristics of Corporate Bonds

1. Restrictive Covenants

- ✓ The bondholders place **some limits** (covenants) on what the company can do (bondholders sort of acting like shareholders)
- ✓ May limit dividends, new debt, mergers, etc.

Q: if covenants are more restrictive, does $i \uparrow$ or $\downarrow$?

Characteristics of Corporate Bonds

1. Restrictive Covenants

✓ The bondholders place **some limits** (covenants) on what the company can do (bondholders sort of acting like shareholders)

✓ May limit <u>dividends</u>, <u>new debt</u>, <u>mergers</u>, etc.

Q: if covenants are more restrictive, does $i\uparrow$ or $\downarrow$?

A: $i\downarrow$. Firm already accepting lots of conditions from bondholders so can lower i $(or\ i_{restr} < i_{oth})$. Also safer for bond holders, as actions against their interests is reduced (lower risk!)

Characteristics of Corporate Bonds

2. Call Provisions

- ✓ A bond may be **callable**
  - issuer can force bondholder to sell back bonds …
    - typically, at par or slightly higher
    - after a "waiting period"

- ✓ All else equal, i for a callable bond is higher than for non-callable bond ($i_{call} > i_{oth}$):
  - because the holder is less able to choose what to do with bond

Q: if $i\downarrow$ is the firm more or less likely to buy back?

A:

Q: Why?

A:

Characteristics of Corporate Bonds

2. Call Provisions

- ✓ A bond may be **callable**
  - issuer can force bondholder to sell back bonds …
    - typically, at par or slightly higher
    - after a "waiting period"
- ✓ All else equal, i for a callable bond is higher than for non-callable bond ($i_{call} > i_{oth}$):
  - because the holder is less able to choose what to do with bond

Q: if $i \downarrow$ is the firm more or less likely to buy back?

A: **more** likely to buy back

Q: Why?

A: Because low *i*:

- allows the firm to re-borrow at a more beneficial rate (lower *i*)
- if *i* low enough, P(market)> P(callable), firm can earn difference

Characteristics of Corporate Bonds

3. Conversion option

- ✓ Debt that ***may*** be converted to **equity** by <u>holder</u>

- ✓ Similar to buying a bond with a *stock option …*
  - opt to convert to equity (see later lecture)
  - … but usually more limited

- ✓ i will be ***lower*** for a convertible bond than for a non-convertible bond $(i_{conv} < i_{oth})$:
  - because the holder is more able to choose what to do with bond (can choose to convert to stock)

- ✓ Firms use it
  - to issue stock, without sending a bad signal to market (of *overvaluation*)
  - they are a cheaper way to raise money

Characteristics of Corporate Bonds

4. Seniority

- ✓ It matters if a company defaults … as firm assets become less valuable

- ✓ *Bond*holders are **senior** to (*i.e.* get paid before) *stock*holders (*i.e.* get paid after)

- ✓ Among bondholders, there is seniority too:
  - ***senior*** bondholders get paid before ***junior*** **(***subordinated***)** bondholders (who are in turn senior to stockholders)

Seniority scheme:

1st senior bondholders

2nd junior bondholders

3rd stockholders

Characteristics of Corporate Bonds

4bis. Collateral

✓ also matters if a company defaults

✓ **Secured** bonds: bond with specific asset pledged as collateral

- these get paid for sure (senior to all)
- ex.: mortgage bonds

✓ **Unsecured** bonds: w/o collateral

- *Debentures*: long-term unsecured bonds
- *Subordinated debentures*: debentures with a lower priority claim

Q: does an **unsecured** bond have higher or lower i compared to secured bonds, all else equal?

Characteristics of Corporate Bonds

4bis. Collateral

✓ also matters if a company defaults

✓ **Secured** bonds: bond with specific asset pledged as collateral

- these get paid for sure (senior to all)
- ex.: mortgage bonds

✓ **Unsecured** bonds: w/o collateral

- *Debentures*: long-term unsecured bonds
- *Subordinated debentures*: debentures with a lower priority claim

Q: does an **unsecured** bond have higher or lower i compared to secured bonds, all else equal?

A: clearly riskier, so i higher, because it lacks guarantee

Bond Pricing: Semi-annual Coupon

So far, we have always assumed annual coupon payments. But actually, most bonds pay coupons semi-annually, so

$$P = \sum_{t=1}^{2n} \frac{C/2}{(1+i/2)^{t}} + \frac{FV}{(1+i/2)^{2n}}$$

- where C, i are annual and n are the years to maturity
- t is a half-year counter

Q: Consider a two-year coupon bond (semi-annual coupon payments) with a face value of $1,000, a coupon rate of 10% and that the interest rate at which investors discount cash flows on similar bonds is 12%? What is the price?

Bond Pricing: Semi-annual Coupon

So far, we have always assumed annual coupon payments. But actually, most bonds pay coupons semi-annually, so

$$P = \sum_{t=1}^{2n} \frac{C/2}{(1+i/2)^{t}} + \frac{FV}{(1+i/2)^{2n}}$$

- where C, i are annual and n are the years to maturity
- t is a half-year counter

Q: Consider a two-year coupon bond (semi-annual coupon payments) with a face value of $1,000, a coupon rate of 10% and that the interest rate at which investors discount cash flows on similar bonds is 12%? What is the price?

A: $\frac{C}{2} = \$50; \frac{i}{2} = 6\%; n = 2; FV = \$1{,}000$

$\frac{C}{2} = \$, 1000 \cdot 10\% \cdot \frac{1}{2}$ *because recall the coupon rate is:* $i_{c.r.} = \frac{C}{FV}$, so $C = i_{c.r.} \cdot FV$

$$P = \frac{\$50}{1+0.06} + \frac{\$50}{(1+0.06)^2} + \frac{\$50}{(1+0.06)^3} + \frac{\$50}{(1+0.06)^4} + \frac{\$1{,}000}{(1+0.06)^4} = \$965.35$$

Bond Pricing: Semi-annual Coupon

- ✓ Why does semi-annual payments make a difference?
- ✓ Because of **compounding**:
  - if you re-invest at shorter intervals, make more money: <u>earn interest on interest already earned (at middle year!)</u>

Example: $1,000 with annual interest of 10% become:

$$\$1{,}000 \times 1.1 = \mathbf{\$1{,}100}$$

With semi-annual payment (5%)

$$\$1{,}000 \times 1.05 \times 1.05 = \mathbf{\$1{,}102.5}$$

Government bonds rating (2017)

	G.Debt (% of GDP)	S&P	Moody's	Fitch
Germany	64.8	AAA	Aaa	AAA
Netherlands	57.6	AAA	Aaa	AAA
Luxembourg	23.0	AAA	Aaa	AAA
Austria	79.8	AA+	Aa1	AA+
Finland	60.8	AA+	Aa1	AA+
France	97.9	AA	Aa2	AA
Belgium	104.5	AA	Aa3	AA−
Estonia	8.6	AA−	A1	A+
Slovakia	52.3	A+	A2	A+
Ireland	71.8	A+	A3	A
Lithuania	40.6	A−	A3	A−
Spain	98.2	BBB+	Baa2	BBB+
Latvia	38.7	A−	A3	A−
Italy	133.5	BBB−	Baa2	BBB
Malta	53.0	A−	A3	A
Slovenia	77.4	A+	Baa3	A−
Portugal	129.5	BB+	Ba1	BB+
Cyprus	103.0	BB+	B1	BB−
Greece	175.0	B−	Caa2	CCC

For a more recent list, see: https://tradingeconomics.com/country-list/rating
or https://www.worldgovernmentbonds.com/world-credit-ratings

US public debt downgraded (optional)

- ✓ US Treasuries are still the safest haven?
  - 2 years ago a first downgrading of US Federal Gov Debt by Fitch!

FINANCIAL TIMES

WORLD US COMPANIES TECH MARKETS CLIMATE OPINION LEX WORK & CAREERS LIFE & ARTS HTSI

FT Alphaville US budget + Add to myFT

Fitch downgrades US on 'erosion of governance'

Fair enough!

  - more recently also other credit rating agencies …
    - ex.: Moodys

MOODY'S RATINGS | Ratings & Regulatory

Ratings News Ratings Search Policies & Codes of Conduct Regulatory Affairs Ratings Definitions Rating Methodologies

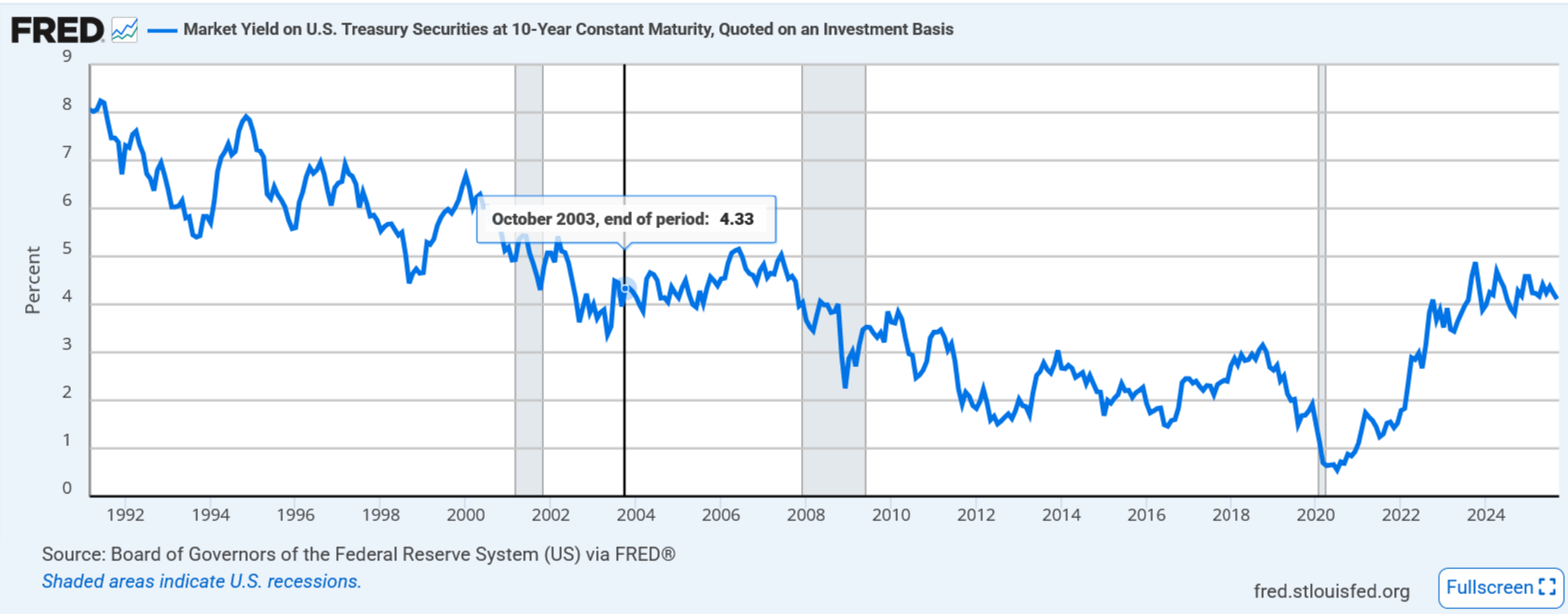
Rating Action

Moody's Ratings downgrades United States ratings to Aa1 from Aaa; changes outlook to stable

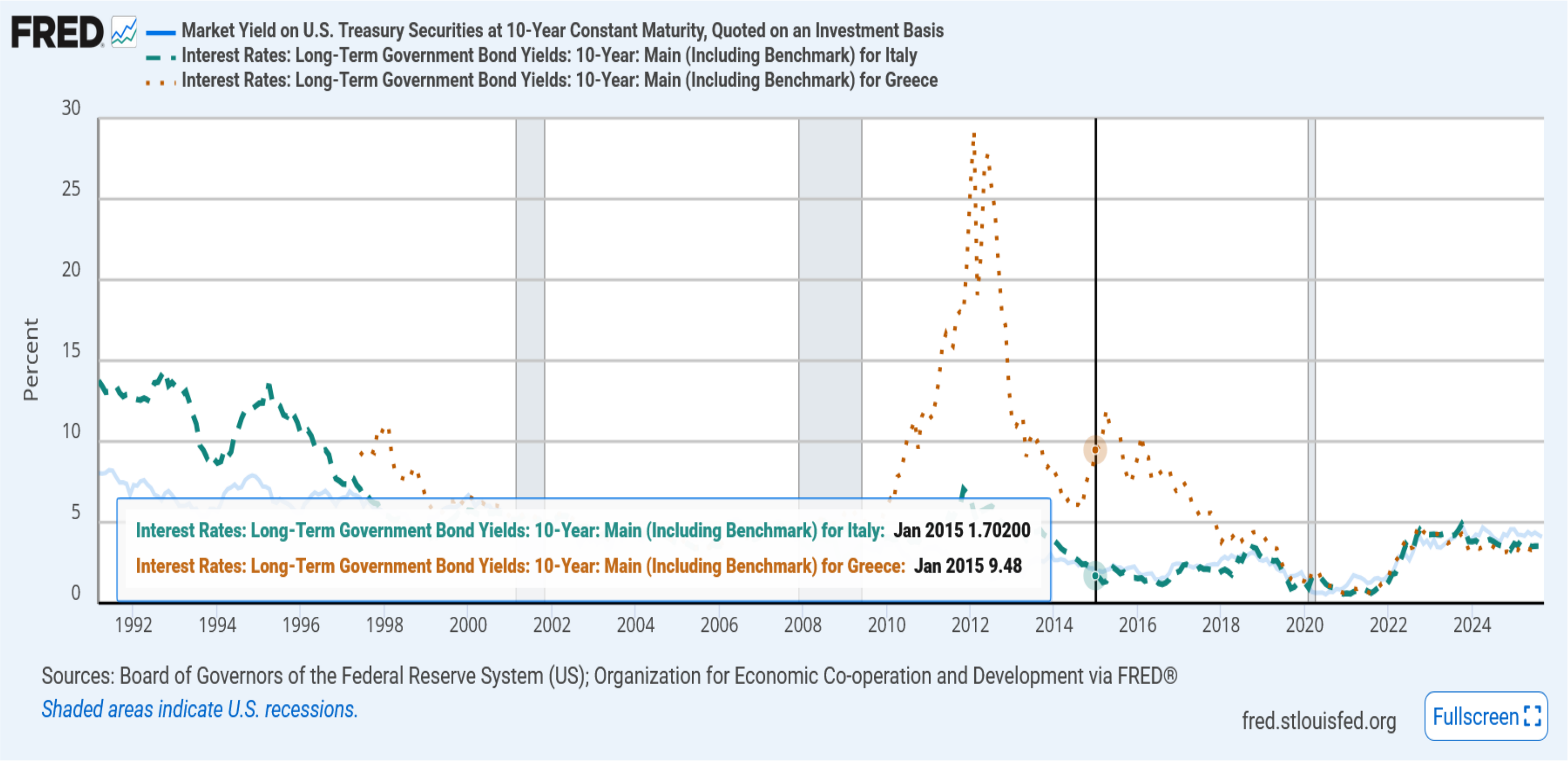
New York, May 16, 2025 -- Moody's Ratings (Moody's) has downgraded the Government of United States of America's (US) long-term issuer and senior unsecured ratings to Aa1 from Aaa and changed the outlook to stable from negative.

  - Interesting reading on VOX

Government bonds interest rates: USA



Government bonds interest rates: USA, ITA, GRC



Foreign holders of US public debt

✓ Optional: who owns US public debt?

Table. **Share of US public debt held by foreign residents**

country	2025-06	2025-03	2024-12	2024-09	2024-06
Japan	12.6%	12.5%	12.3%	12.5%	13.2%
China (Mainland + Hong Kong)	**10.9%**	**11.4%**	**11.8%**	**11.6%**	**12.1%**
United Kingdom	9.4%	8.6%	8.4%	8.8%	9.0%
Cayman Islands	4.9%	5.0%	4.9%	4.8%	3.9%
Canada	4.8%	4.7%	4.4%	4.2%	4.5%
Belgium	4.7%	4.4%	4.3%	4.2%	3.8%
Luxembourg	4.4%	4.6%	4.9%	4.6%	4.5%
France	4.1%	4.0%	3.9%	3.8%	3.7%
Ireland	3.5%	3.6%	3.9%	4.1%	4.0%
Switzerland	3.3%	3.4%	3.5%	3.5%	3.5%
Taiwan	**3.3%**	**3.3%**	**3.3%**	**3.3%**	**3.2%**

Source: US Treasury

✓and … only about 30% of US public debt is held by foreign residents (Congress data)